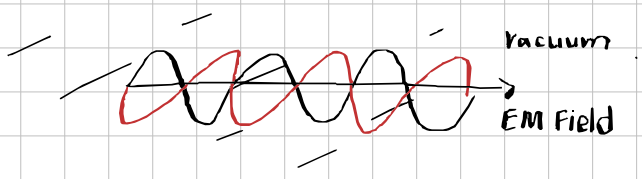
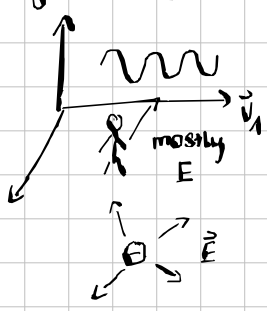
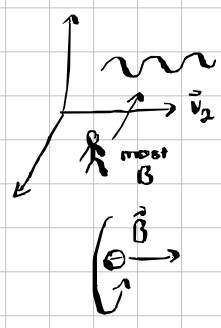
Electromagnetic Pieces: E and B

Think of light as a chess game.
Vacuum is the board on which electromagnetic field exists.



Usually, it is easy to think of electromagnetic field as composed Electric and Magnetic field. This thinking is technically correct, but it is not the full story. We will understand Electric and magnetic field first. $\vec{E}$ and $\vec{B}$ are not two seperate things. They are two aspects of a single electromagnetic field.





Reference changes what one see

At a fixed reference, they are usually thought as "position" and "momentum". Well, at quantum level,

$$\hat{X} = \frac{\hat{a} + \hat{a}^\dagger}{\sqrt{2}} \quad \hat{P} = \frac{\hat{a} - \hat{a}^\dagger}{i\sqrt{2}} \quad [\hat{X}, \hat{P}] = i \quad \text{and light as } |\psi\rangle$$

Where $\hat{X}$ and $\hat{P}$ are measuring the phase of $|\psi\rangle$ [refer to]
For some cases of special reference choice

$$\hat{E} \propto \hat{X} \quad \text{and} \quad \hat{B} \propto \hat{P} \quad \rightarrow \text{However, } \vec{E} \text{ and } \hat{B} \text{ can be } \alpha \propto \hat{X} + \beta \hat{P}$$

The results of $\hat{X}$ and $\hat{P}$ can represent $\hat{E}$ and $\hat{B}$. This means $\hat{E}$ and $\hat{B}$ are not compatible.
They might be compatible where the measured location is different.

$$[\hat{E}(\vec{r}), \hat{B}(\vec{r}')] \propto \delta(\vec{r} - \vec{r})$$

But, locally they are not
Well, then why we say E and B can be found both at the same time?
Take coherent light for example,



When amplitude becomes strong, the uncertainty becomes less important. That is why we can know both $\vec{E}$ and $\vec{B}$.

ε_0 and μ_0

Now, how E and B move in vacuum scales as ε_0 and μ_0. Together, they determine how EM waves propagate through.

$$\varepsilon_0 = 8.854e-12 \ F/m \qquad F = S/\Omega$$
$$\mu_0 = 4\pi e-7 \ H/m \qquad H = S\Omega$$

Because

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad \text{and} \quad \nabla \times \vec{B} = \mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t}$$

ε_0 and μ_0 become entangled to each other. The coupling feedback controlling $E \to B \to E \to B$ determine how quickly disturbances move

$$\nabla^2 \vec{E} = \mu_0 \varepsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2} \equiv \nabla^2 \psi = \frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2} \qquad c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}$$

Now, if we think of scaling of how E and B through vacuum, For a plane wave

$$B = \frac{E}{c} \qquad H = \frac{B}{\mu_0} \Rightarrow \frac{E}{H} = \mu_0 c = \sqrt{\frac{\mu_0}{\varepsilon_0}} = Z_0 = 377\Omega$$

The impedance Z_0 sets the relative between E and H.
ε_0 and μ_0 together tell how fast the electromagnetic character moves through coupling and how its electric and magnetics are relative through scale.
Two concepts not to be conflated:

$E = Z_0 H$ relates electric and magnetic amplitudes, but units are different

One should never conclude E is much bigger than H. In fact, their energy is the same.

$$\frac{1}{2}\varepsilon_0 E^2 = \frac{1}{2}\mu_0 H^2$$

What people usually mean when "magnetic part can be ignored" is when matter couples to E field than H field.

$$\vec{F} = q(\vec{E} + \vec{v}\vec{B}) \qquad B \sim E/c \qquad \frac{F_B}{F_E} \sim \frac{vB}{E} \sim \frac{v}{c}$$

For electrons bound in ordinary atoms,

$$v \ll c \Rightarrow F_B \ll F_E$$

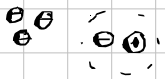
Four Main Material Sources

There are four main material source quantities in macroscopic electromagnetism

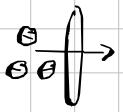
$$\rho_{free}, \vec{J}_{free}, \vec{P}, \vec{M}$$

They are all continuous functions of space $\vec{r}$ and time t.

- Free charge density: ρ_{free} C/m^3 $C = A \cdot s$



- Free current density $\vec{J}_{free}$ A/m^2 measures how much charge flows through a cross sectional area



- Polarization:

$\vec{P}$ C/m^2

- Magnetization

$\vec{M}$ A/m

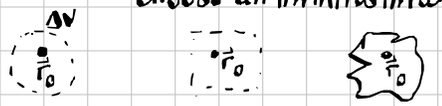
The current density $\vec{J}$ could arise from the motion of ρ such as $\vec{J} = \rho\vec{v}$ where $\vec{v}$ is the velocity of the charge



J

Rules: mathematics

Divergence: it acts on a vector field and output a scalar field. To determine, we freeze in time $t = t_0$ choose an infinitesimal closed surface-volume ΔV around $\vec{r}_0$. The shape is irrelevant.



Now, we divide closed surface into multiple elements Δs with direction pointing outward.



Divergence is then defined as

$$\nabla \cdot \vec{V}(r_0, t_0) = \lim_{\Delta V \to 0} \frac{\oint \vec{V}(r, t_0) \cdot d\vec{s}}{\Delta V}$$

$\oint \vec{V}(r, t_0) \cdot d\vec{s}$ is summing up all the contribution of $\vec{V}$ outward

ΔV avoids the integral going to zero.

This geometrical version can be rewritten as

$$\nabla \cdot \vec{V} = \frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z}$$

This concept is great at describing whether something is a source or a sink

Theorem of Gauss: consider an arbitrary volume, the divergence's definition relates to a very significant theorem called theorem of Gauss

$$\nabla \cdot \vec{V}(r_0, t_0) = \lim_{\Delta V \to 0} \frac{\oint \vec{V}(r, t_0) \cdot d\vec{s}}{\Delta V} \Rightarrow \int_{volume} \nabla \cdot \vec{V}(r, t)\, dv = \oint_{surface} \vec{V}(r, t) \cdot d\vec{s}$$

The value of V on the surface could be found by knowing how much sources are inside the arbitrary volume. The left side states that if the two adjacent volumes have the surfaces in contact to be canceled out due to different orientations. The only part that does not cancel is the boundary which is the right side.

Curl: produces another vector to describe how curl the original vector is. To define the curl,



Freeze in time $t = t_0$, draw an infinitesimal flat loop around $\vec{r}_0$ with $\vec{r}_0$ in the middle. Right hand rule to choose the direction for $\hat{n}$ and $d\ell$



$$\hat{n} \cdot [\nabla \times \vec{V}(r_0, t_0)] = \lim_{\Delta s \to 0} \frac{1}{\Delta S} \oint_{loop} \vec{V}(r, t_0) \cdot d\vec{\ell}$$

$$\nabla \times V = \left(\frac{\partial V_z}{\partial y} - \frac{\partial V_y}{\partial z}\right)\hat{x} + \left(\frac{\partial V_x}{\partial z} - \frac{\partial V_z}{\partial x}\right)\hat{y} + \left(\frac{\partial V_y}{\partial x} - \frac{\partial V_x}{\partial y}\right)\hat{z}$$

Theorem of Stokes: we divide surface into a large number of small surfaces. This is the same as the sum of all infinitesimal small surfaces. When small surfaces are in contact, they cancel. Those without adjacent neighbors survive.



$$\int_{surface} [\nabla \times \vec{V}(r, t)] \cdot d\vec{s} = \oint_{boundary} \vec{V}(r, t) \cdot d\vec{\ell}$$

The shape is irrelavant due to the reason above. For a closed surface, there will no longer be a boundary.

$$\nabla \cdot \nabla \times \vec{V} = 0$$